

HDOC Day-19 assignment

AIM :

Write a c program on the algorithm for calculating the semester grade point average (SGPA) and Cumulative Grade Point Average (CGPA) for 'm' semesters. Each semester has 'n' courses, where 'n' may vary between semesters. Assume that the student has passed all courses.

ALGORITHM :

1. Start.
2. Define the variables and their types: global_id, m, n, semester, course, code, credit, GP, semester_TC, semester_TGC, cumulative_TC, and cumulative_TGC as integers; grade as a character; and SGPA and CGPA as floating-point variables.
3. Get global_id.
4. Validate global_id: If global_id is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
5. Get m.
6. Validate m: If m is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
7. Set the cumulative values: cumulative_TC = 0, cumulative_TGC = 0
8. Repeat Steps 8.1 to 8.8 for m semesters:
 - 8.1. Set the semester totals: semester_TC = 0, semester_TGC = 0
 - 8.2. Get n.
 - 8.3. Validate n: If n is not positive, display an appropriate error message and request it again. Repeat until a positive value is entered.
 - 8.4. Repeat Steps 8.4.1 to 8.4.9 for n courses:

8.4.1. Get code.

8.4.2. Validate code: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.

8.4.3. Get credit.

8.4.4. Validate credit: If it is not positive, display an error message and request it again. Repeat until a positive value is entered.

8.4.5. Get grade.

8.4.6. Validate grade: Accept only O, A, B, C, D, E, or P. If the grade is invalid, display an error message and request it again. Repeat until a valid grade is entered.

8.4.7. Find GP corresponding to grade using the university's conversion scale:
O = 10, A = 9, B = 8, C = 7, D = 6, E = 5, P = 4

8.4.8. Calculate the semester total grade credits: $\text{semester_TGC} = \text{semester_TGC} + (\text{GP} * \text{credit})$

8.4.9. Calculate the semester total credits: $\text{semester_TC} = \text{semester_TC} + \text{credit}$

8.5. Calculate the SGPA of the current semester: $\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$

8.6. Display semester and SGPA.

8.7. Add the semester grade credits to the cumulative grade credits:
 $\text{cumulative_TGC} = \text{cumulative_TGC} + \text{semester_TGC}$

8.8. Add the semester credits to the cumulative credits: $\text{cumulative_TC} = \text{cumulative_TC} + \text{semester_TC}$

9. Calculate the CGPA: $CGPA = \text{cumulative_TGC} / \text{cumulative_TC}$

10. Display global_id, m, cumulative_TC, and CGPA.

11. Stop

C PROGRAM IMPLEMENTATION :

```
UW PICU 5.07 file: 170
#include <stdio.h>

int main() {
    int global_id, m, n, semester, course, code, credit, GP;
    int semester_TC, semester_TGC;
    int cumulative_TC = 0, cumulative_TGC = 0;
    char grade;
    float SGPA, CGPA;

    printf("Enter Global ID: ");
    scanf("%d", &global_id);
    while (global_id <= 0) {
        printf("Invalid ID. Enter positive Global ID: ");
        scanf("%d", &global_id);
    }

    printf("Enter number of semesters: ");
    scanf("%d", &m);
    while (m <= 0) {
        printf("Invalid number. Enter positive semesters: ");
        scanf("%d", &m);
    }

    for (semester = 1; semester <= m; semester++) {
        printf("\n--- Semester %d ---\n", semester);

        semester_TC = 0;
        semester_TGC = 0;

        printf("Enter number of courses: ");
        scanf("%d", &n);
        while (n <= 0) {
            printf("Invalid number. Enter positive courses: ");
            scanf("%d", &n);
        }

        for (course = 1; course <= n; course++) {
            printf("\nCourse %d:\n", course);

            printf("Enter course code: ");
            scanf("%d", &code);
            while (code <= 0) {
                printf("Invalid code. Enter positive course code: ");
                scanf("%d", &code);
            }

            printf("Enter credit: ");
            scanf("%d", &credit);
            while (credit <= 0) {
                printf("Invalid credit. Enter positive credit: ");
                scanf("%d", &credit);
            }
        }
    }
}
```

```

while (1) {
    printf("Enter grade (O, A, B, C, D, E, P): ");
    scanf(" %c", &grade);

    if (grade == 'O' || grade == 'o') { GP = 10; break; }
    else if (grade == 'A' || grade == 'a') { GP = 9; break; }
    else if (grade == 'B' || grade == 'b') { GP = 8; break; }
    else if (grade == 'C' || grade == 'c') { GP = 7; break; }
    else if (grade == 'D' || grade == 'd') { GP = 6; break; }
    else if (grade == 'E' || grade == 'e') { GP = 5; break; }
    else if (grade == 'P' || grade == 'p') { GP = 4; break; }
    else {
        printf("Invalid grade! Try again.\n");
    }
}

semester_TGC = semester_TGC + (GP * credit);
semester_TC = semester_TC + credit;
}

SGPA = (float)semester_TGC / semester_TC;
printf("\nSemester %d SGPA = %.2f\n", semester, SGPA);

cumulative_TGC = cumulative_TGC + semester_TGC;
cumulative_TC = cumulative_TC + semester_TC;
}

```

```

CGPA = (float)cumulative_TGC / cumulative_TC;

printf("\n-----\n");
printf("Global ID: %d\n", global_id);
printf("Total Semesters: %d\n", m);
printf("Total Credits: %d\n", cumulative_TC);
printf("Final CGPA: %.2f\n", CGPA);
printf("-----\n");

return 0;
}

```

OUTPUT SCREENSHOT :

```
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % nano 19q1.c
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % gcc 19q1.c -o 19q1
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./19q1
Enter Global ID: 10261
Enter number of semesters: 1

--- Semester 1 ---
Enter number of courses: 2

Course 1:
Enter course code: 101
Enter credit: 4
Enter grade (O, A, B, C, D, E, P): A

Course 2:
Enter course code: 202
Enter credit: 4
Enter grade (O, A, B, C, D, E, P): B

Semester 1 SGPA = 8.50

-----
Global ID: 10261
Total Semesters: 1
Total Credits: 8
Final CGPA: 8.50
-----
[manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % ./19q1
Enter Global ID: 10262
Enter number of semesters: 1

--- Semester 1 ---
Enter number of courses: 3

Course 1:
Enter course code: 287
Enter credit: 3
Enter grade (O, A, B, C, D, E, P): B

Course 2:
Enter course code: 890
Enter credit: 4
Enter grade (O, A, B, C, D, E, P): B

Course 3:
Enter course code: 126
Enter credit: 3
Enter grade (O, A, B, C, D, E, P): A

Semester 1 SGPA = 8.30

-----
Global ID: 10262
Total Semesters: 1
Total Credits: 10
Final CGPA: 8.30
-----
manalithakar@Unknown_4e:f1:c3:0c:fe:82 ~ % █
```